MZ UCA Cloud Academy

AWS Solutions Architect Associate (SAA)

Architecture Defense 🛡️

AWS Organizations, Service Control Policies (SCPs), Landing Zones & Multi-Account Strategy

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Cloud Governance Architect

◈ PURPOSE OF THE ARCHITECTURE DEFENSE

In the real world, architects do not simply build solutions.

They defend their decisions.

Senior architects are expected to justify:

✔ Security decisions

✔ Governance controls

✔ Cost management approaches

✔ Compliance strategies

✔ Risk mitigation measures

✔ Scalability plans

Your Mission

You are the Lead Cloud Governance Architect for a rapidly growing enterprise.

You must evaluate the current environment and defend your proposed architecture before an Executive Review Board.

◈ BUSINESS CASE

Ubuntu Retail Group (URG)

URG is expanding across Southern Africa and migrating critical business systems to AWS.

Business Requirements

✔ Secure cloud environment

✔ Regulatory compliance

✔ Cost transparency

✔ High availability

✔ Centralized governance

✔ Scalable cloud operating model

✔ Enterprise-grade security

Current Environment

Production

Development

Testing

Security

Finance

All operate within a single AWS account.

Executive Concern

"Can this architecture support long-term growth safely?"

◈ DEFENSE ROUND 1

SHOULD WE USE AWS ORGANIZATIONS?

Executive Question

"Our engineers already have an AWS account.

Why do we need AWS Organizations?"

Your Defense

Explain:

1. What AWS Organizations provides.

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1. What governance problems it solves.

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1. What risks exist if we remain on a single account.

Architect Recommendation

☐ Implement AWS Organizations

☐ Remain Single Account

Defend Your Choice

◈ DEFENSE ROUND 2

DESIGNING ORGANIZATIONAL UNITS

Executive Question

"Why can't we simply place all accounts together?"

Task

Design your Organizational Unit structure.

AWS Organization

│

├── ______

│

├── ______

│

├── ______

│

├── ______

│

└───── ─────

Defense Questions

Why was each OU created?

How does it improve governance?

Architecture Principle

Structure should reflect governance requirements.

◈ DEFENSE ROUND 3

SCP GOVERNANCE STRATEGY

Executive Question

"Why do we need Service Control Policies if IAM already exists?"

Explain

Difference between:

IAM Policies

and

SCPs

IAM Policies:

SCPs:

Governance Defense

Provide three examples where SCPs protect the organization.

1.

2.

3.

Critical Principle

SCPs establish organizational guardrails.

◈ DEFENSE ROUND 4

LANDING ZONE INVESTMENT

Executive Question

"Why should we spend time building a Landing Zone before deploying workloads?"

Your Defense

What is a Landing Zone?

What capabilities does it provide?

How does it reduce future risk?

Architecture Principle

Strong foundations reduce future complexity.

◈ DEFENSE ROUND 5

MULTI-ACCOUNT STRATEGY

Executive Question

"Why not keep everything inside one account?"

Defend Multi-Account Architecture

Security Benefits

Cost Benefits

Governance Benefits

Compliance Benefits

Executive Summary

Why is account isolation considered an AWS best practice?

◈ DEFENSE ROUND 6

THE BLAST RADIUS CHALLENGE

Scenario

A developer accidentally deletes production resources.

Architecture A

Single Account Model

Impact:

Architecture B

Multi-Account Model

Impact:

Defense Question

Which architecture better limits business risk?

Why?

Architecture Principle

Great architects design for failure before failure occurs.

◈ DEFENSE ROUND 7

COST GOVERNANCE

Executive Question

"How do we maintain financial control across multiple accounts?"

Your Defense

Which AWS Organizations feature helps solve this challenge?

How does it improve visibility?

How does it improve governance?

Financial Architecture Principle

Visibility creates accountability.

◈ DEFENSE ROUND 8

SECURITY GOVERNANCE

Scenario

Developers request permission to:

✔ Disable CloudTrail

✔ Disable GuardDuty

✔ Remove Security Hub

Executive Question

Should these permissions be approved?

☐ YES

☐ NO

Defend Your Position

Governance Recommendation

Which SCP controls would you implement?

◈ DEFENSE ROUND 9

ENTERPRISE ACCOUNT MODEL

Executive Board Request

Design the AWS account structure you would recommend.

Your Architecture

AWS Organization

```
│
├── ____
├── ____
├── ____
├── ____
├── ____
├── ____
└── ____
```

Defense

Explain why each account exists.

◈ DEFENSE ROUND 10

EXECUTIVE ARCHITECT REVIEW

You are presenting to:

◆ Chief Executive Officer

◆ Chief Technology Officer

◆ Chief Information Security Officer

◆ Chief Financial Officer

Final Executive Questions

1. Why is governance essential in AWS?

1. Why should AWS Organizations be implemented?

1. Why are SCPs critical?

1. Why should workloads be separated?

1. Why is a Landing Zone valuable?

◈ ARCHITECT DECISION MATRIX

Rate your proposed architecture.

Category	Score (1–5)
Security	_____
Governance	_____
Compliance	_____
Cost Management	_____
Scalability	_____
Resilience	_____
Operational Excellence	_____

Total Score

__ / 35

Architecture Rating

30–35

✔ Enterprise Architecture Ready

24–29

✔ Strong Architecture

18–23

✔ Moderate Maturity

Below 18

✔ Architecture Requires Improvement

◈ EXECUTIVE APPROVAL RECOMMENDATION

As the Lead Cloud Governance Architect:

Would you approve this architecture for enterprise deployment?

☐ YES

☐ NO

Executive Summary

◈ THINK LIKE A CLOUD GOVERNANCE ARCHITECT

Before approving any cloud environment ask:

◆ Is governance centralized?

◆ Are security controls protected?

◆ Is blast radius minimized?

◆ Is compliance enforceable?

◆ Can the environment scale safely?

◆ Are financial controls visible?

◆ Is operational ownership clear?

◈ FINAL TAKEAWAY

Technology alone does not create enterprise readiness.

Governance transforms cloud infrastructure into a secure, scalable, and sustainable operating model.

The role of a Cloud Governance Architect is not merely to deploy resources.

It is to ensure that every resource exists within a framework of accountability, control, and trust.

MZ UCA ARCHITECT PRINCIPLE

"Anyone can deploy cloud resources.

Architects design systems.

Governance Architects design control."

Zulu Mthulisi

Cloud Governance Architect

MZ UCA Cloud Academy

Building Architects. Establishing Governance. Creating Leaders.